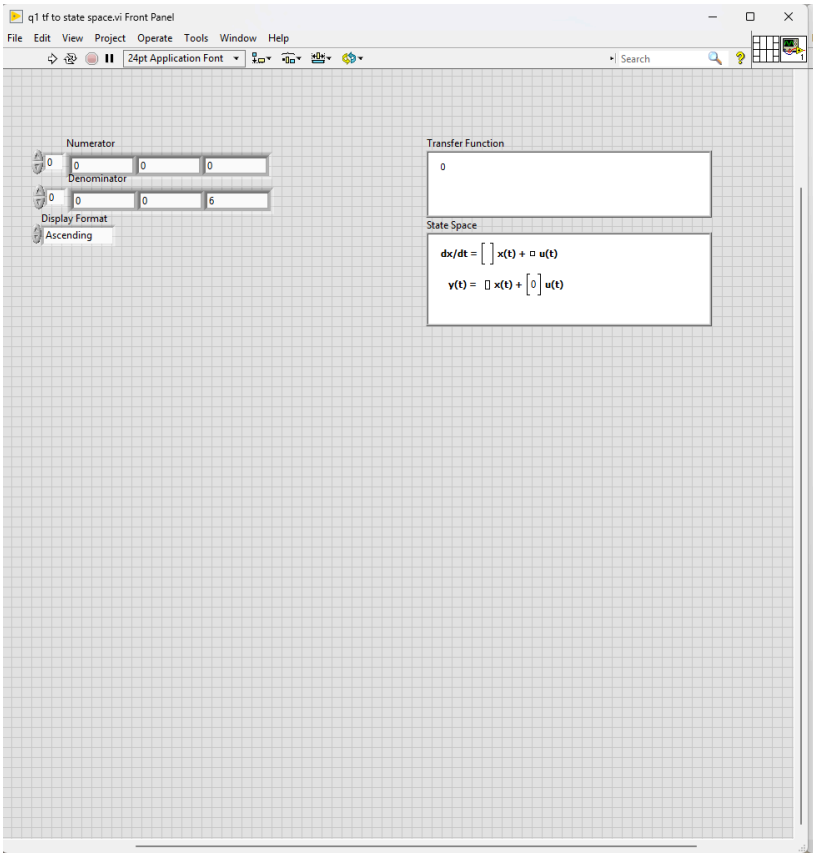
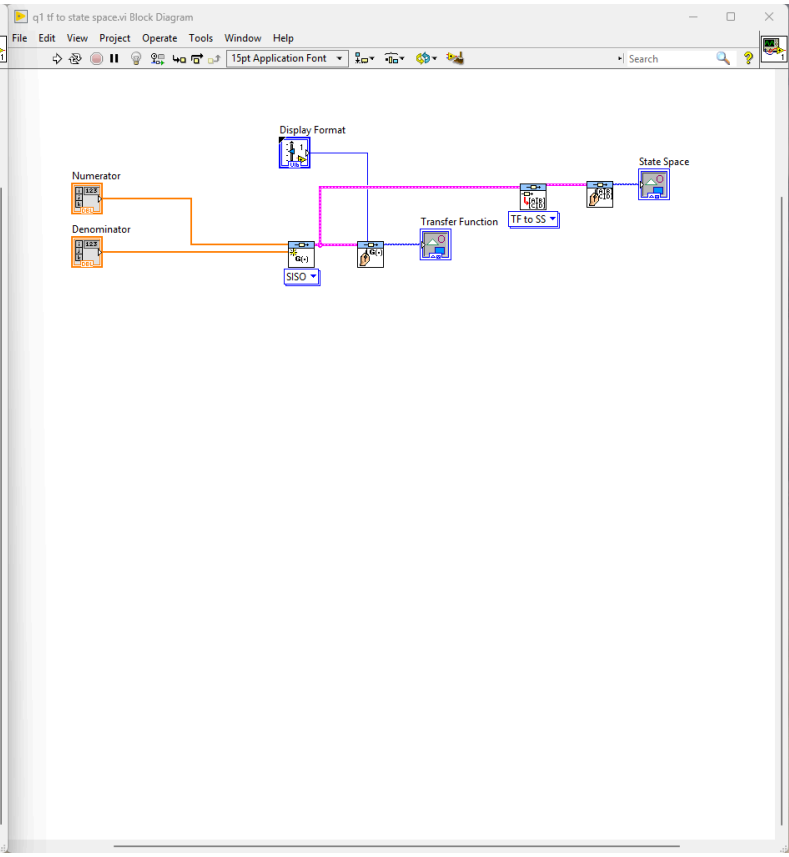
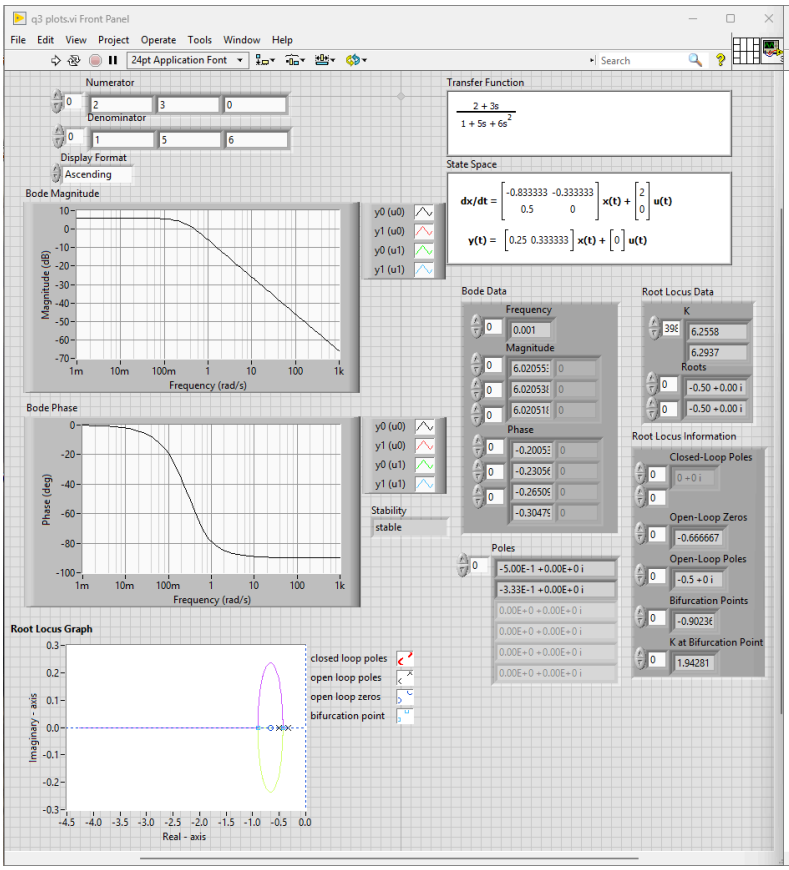
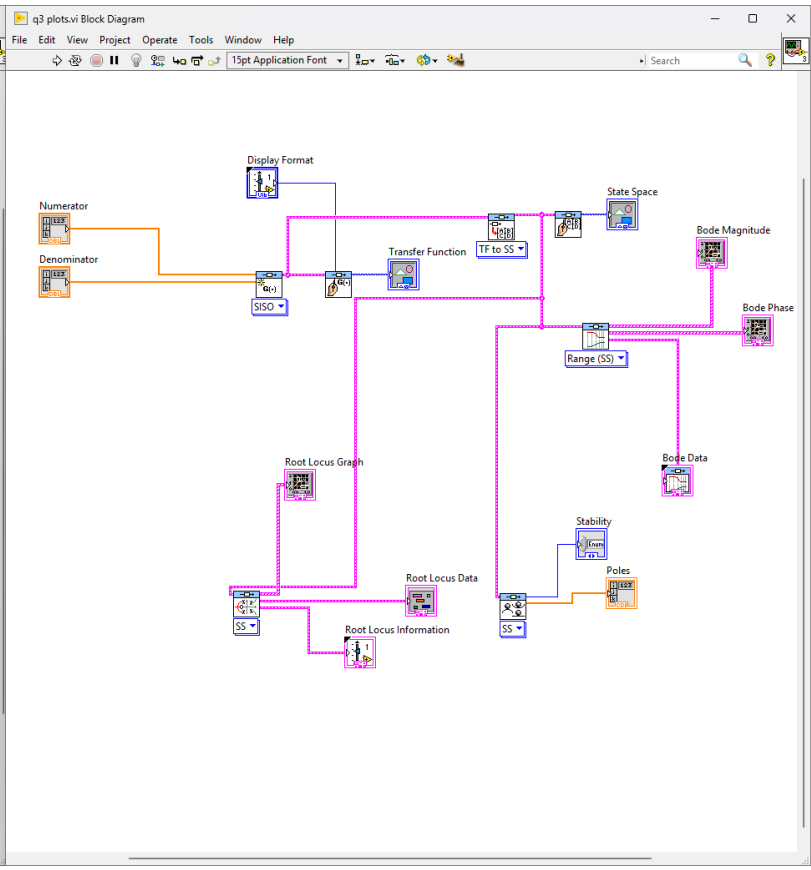
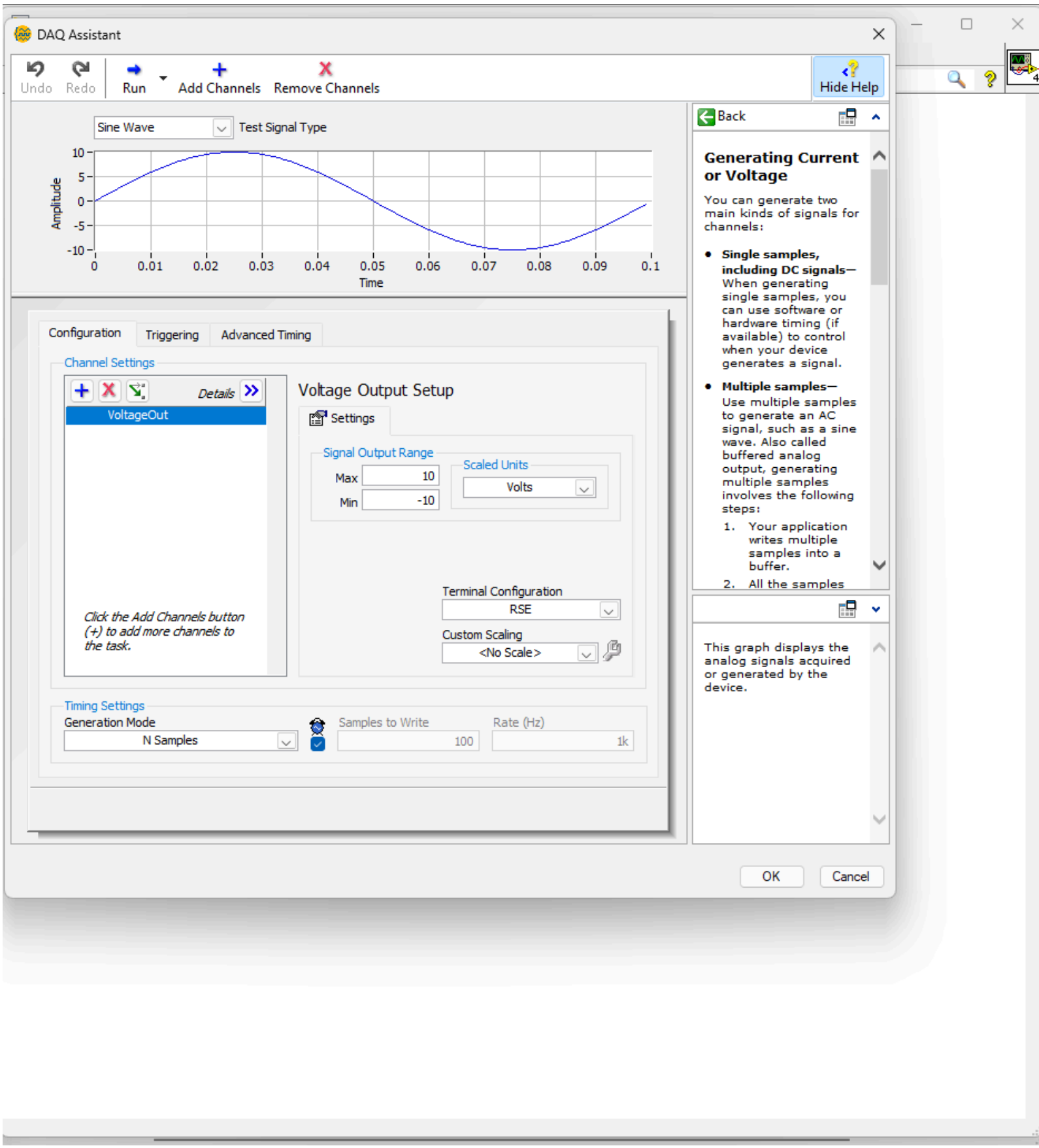
Lab 10: Control and simulation toolbox

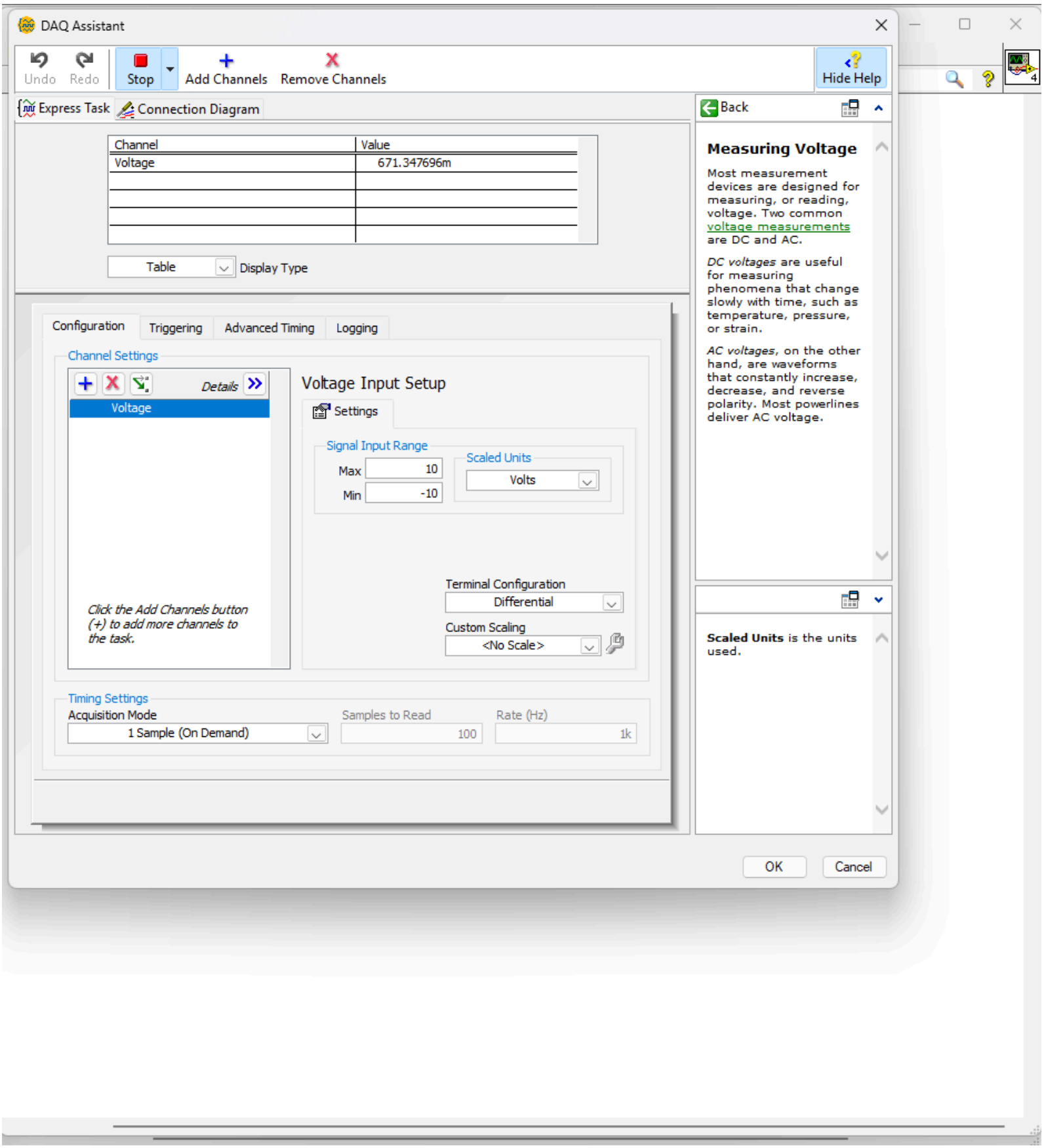


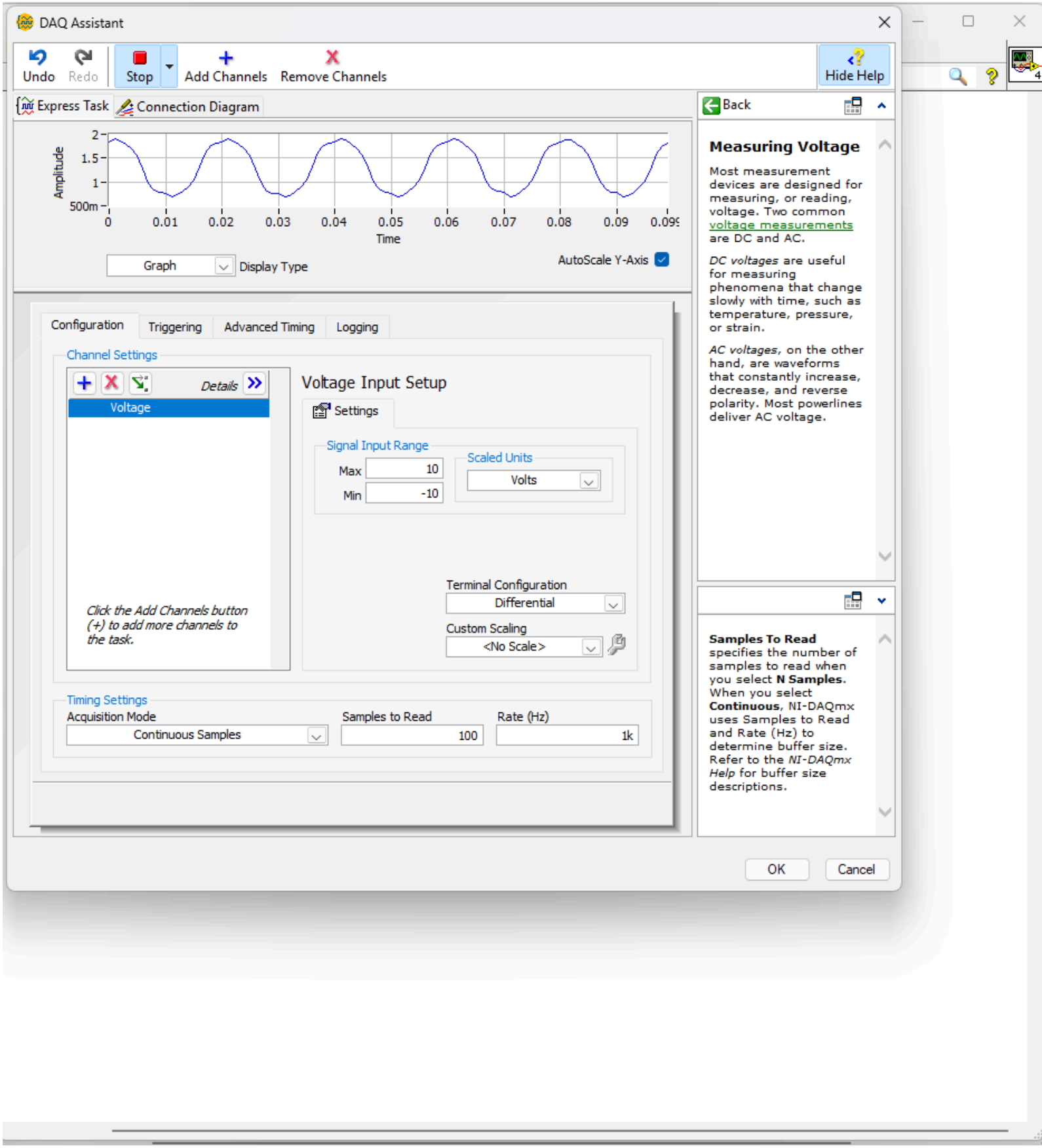







DAQ Assistant
Undo
Redo
Run
Add Channels
Remove Channels
Hide Help
Sine Wave
Test Signal Type
Amplitude
Time
Configuration
Triggering
Advanced Timing
Channel Settings
Details
VoltageOut
Click the Add Channels button (+) to add more channels to the task.
Voltage Output Setup
Settings
Signal Output Range
Max
10
Min
-10
Scaled Units
Volts
Terminal Configuration
RSE
Custom Scaling
<No Scale>
Timing Settings
Generation Mode
N Samples
Samples to Write
100
Rate (Hz)
1k
Back
Generating Current or Voltage
You can generate two main kinds of signals for channels:
Single samples, including DC signals—When generating single samples, you can use software or hardware timing (if available) to control when your device generates a signal.
Multiple samples—Use multiple samples to generate an AC signal, such as a sine wave. Also called buffered analog output, generating multiple samples involves the following steps:
1. Your application writes multiple samples into a buffer.
2. All the samples
This graph displays the analog signals acquired or generated by the device.
OK
Cancel


DAQ Assistant
Undo
Redo
Stop
Add Channels
Remove Channels
Hide Help
Express Task
Connection Diagram
Back
Channel
Value
Voltage
671.347696m
Table
Display Type
Measuring Voltage
Most measurement devices are designed for measuring, or reading, voltage. Two common voltage measurements are DC and AC.
DC voltages are useful for measuring phenomena that change slowly with time, such as temperature, pressure, or strain.
AC voltages, on the other hand, are waveforms that constantly increase, decrease, and reverse polarity. Most powerlines deliver AC voltage.
Configuration
Triggering
Advanced Timing
Logging
Channel Settings
Details
Voltage
Click the Add Channels button (+) to add more channels to the task.
Voltage Input Setup
Settings
Signal Input Range
Max
10
Min
-10
Scaled Units
Volts
Terminal Configuration
Differential
Custom Scaling
<No Scale>
Scaled Units is the units used.
Timing Settings
Acquisition Mode
1 Sample (On Demand)
Samples to Read
100
Rate (Hz)
1k
OK
Cancel


DAQ Assistant
Undo
Redo
Stop
Add Channels
Remove Channels
Hide Help
Express Task
Connection Diagram
Back
Amplitude
2
1.5
1
500m
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.09
Time
Graph
Display Type
AutoScale Y-Axis
Measuring Voltage
Most measurement devices are designed for measuring, or reading, voltage. Two common voltage measurements are DC and AC.
DC voltages are useful for measuring phenomena that change slowly with time, such as temperature, pressure, or strain.
AC voltages, on the other hand, are waveforms that constantly increase, decrease, and reverse polarity. Most powerlines deliver AC voltage.
Configuration
Triggering
Advanced Timing
Logging
Channel Settings
Details
Voltage
Click the Add Channels button (+) to add more channels to the task.
Voltage Input Setup
Settings
Signal Input Range
Max
10
Min
-10
Scaled Units
Volts
Terminal Configuration
Differential
Custom Scaling
<No Scale>
Timing Settings
Acquisition Mode
Continuous Samples
Samples to Read
100
Rate (Hz)
1k
Samples To Read specifies the number of samples to read when you select N Samples. When you select Continuous, NI-DAQmx uses Samples to Read and Rate (Hz) to determine buffer size. Refer to the NI-DAQmx Help for buffer size descriptions.
OK
Cancel